LESSON PLAN · SCRATCH JUNIOR (K–2) ADVANCED: GAME BUILDERS

Lesson 9: Project: Relay Race

Module 3 | 35 minutes | Project: messages | KG–Grade 2 · ages 6–7

Objective: Plan and build a relay race where runners pass a message to start the next runner, ending with a winner.

Success criteria

- My runners pass messages.
- Each runner waits for its turn.
- My relay ends with a celebration.

Key vocabulary

- **baton**: What relay runners pass.
- **chain**: One thing starts the next.
- **celebrate**: Show you're happy.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (35 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Class relay (5 min): three teams pass a baton.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
10 min	**Build**	Build-along steps, then Last leg. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; My Relay Race.

Unplugged warm-up

Class relay (5 min): three teams pass a baton.

Teaching the idea

Chained messages coordinate three sprites in sequence; colour channels must match. Assess understanding via the reflection ("how does runner 2 know?").

Analogy: A real relay race with batons.

Live demo: Show a two-runner relay; ask how to add a third.

Questions to ask

Ask	Listen for
What starts runner 3?	The blue message.
What if colours don't match?	Runner never starts.
How does it end?	Celebration.

Differentiation

Support
Two runners only.

Stretch
A fourth runner and a cheering crowd.

Common misconceptions

Misconception	How to fix it
All runners start on the flag	Only the first one.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Messages	Not yet	One message	Two messages chain correctly	Explains how messages work
Sequence	Not yet	Runners move	Runners take turns	Smooth race with a celebration
Testing	Not tested	Tested with help	Found and fixed a problem	Explains the fix

Exit ticket
What starts runner 2?

Homework
Explain your relay to someone at home.

Answer key — Kitabash activities

1. Last leg (perform · block code)

One solution:



2. My Relay Race (probe · project)

Answer key — worksheet

1. **Key words:** baton = What relay runners pass. · chain = One thing starts the next. · celebrate = Show you're happy.
2. **My plan:** A clear plan that covers every requirement.
3. **Checklist:** All requirements met.
4. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.